

Case Study Rubric – Text Analysis of News vs Opinion Headlines

Due: TBD

Submission format: Upload PDF and link to GitHub repository to Canvas

General Description

Submit to Canvas a PDF of your written analysis and a link to your GitHub repository containing code and data. This case study explores how text analysis can be used to compare language patterns between News and Opinion article titles from The Cavalier Daily.

Why am I doing this?

This assignment gives you the opportunity to apply fundamental text analysis techniques to a real-world dataset. You will learn how language patterns differ across contexts and how statistical methods can be used to support conclusions. This mirrors how data science is used to analyze communication, media, and human behavior.

What am I going to do?

You will analyze two sets of article titles (News and Opinion) using a bag-of-words approach. You will compare word frequency distributions and apply a chi-square test to determine whether differences between the two groups are statistically significant.

Your deliverable will include:

A written analysis (PDF) supported by reviewing and interpreting provided data, outputs, and analysis workflow.

Students are not required to write or modify code. Instead, they will interpret results generated from the provided analysis and explain findings in a clear and structured report.

How will I know I have succeeded?

You will meet expectations on this case study when you follow the criteria in the rubric below.

Category	Details
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Formatting	Submit each component listed below: Written Portion • Submit as a PDF file • 1–2 pages • Clear organization and readable formatting References • Include a reference section at the end of the PDF • Use a consistent citation format (e.g., IEEE or APA)
Written Portion	Discuss your interpretation of the study and thought process by addressing the following: • In a short paragraph, summarize the goal of the analysis and why it is meaningful. • In a short paragraph, describe your approach to comparing News and Opinion headlines. ◦ Include a simple visual or diagram showing your analysis process (e.g., text → cleaning → word counts → comparison → test). • Present and explain your results in complete sentences. ◦ Include key findings about word usage differences. ◦ Clearly interpret the results of the chi-square test.
Reflection	• In a short paragraph, describe any challenges encountered during the analysis. • Explain how you addressed those challenges. • Discuss what you would improve if repeating the project.
References	• Include a list of references at the end of the written portion. • References should support your understanding of: ◦ Bag-of-words models

	○ Text preprocessing (e.g., stopwords, tokenization) ○ Chi-square testing ● Use a consistent citation format.
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